

To: Devon Achebe, VP of Customer Retention, Summit Telecom
From: Ian Christiansen
24 September, 2026
Subject: Retention Pilot Recommendation

Hello Devon,

For a limited retention pilot, I recommend that Summit Telecom uses a boosted trees model to select customers. Summit Telecom should start by contacting the 20% of customers with the highest predicted churn risk, rather than immediately starting with a full campaign. Logistic regression should remain available as a strong second alternative, because its performance was very close to that of the boosted trees model.

One reason that I selected boosted trees was because it had the highest validation AUC at 0.846, compared with 0.838 for logistic regression and 0.743 for the contract rule. But this alone cannot be the only indicator in predictive capabilities. Once I ran the final test, it also supported the highest AUC choice of boosted trees. The final test had an AUC slightly higher than the original, at 0.850, and identified about 195 actual churners among the 281 customers on its contact list. This resulted in a top-20% churn rate of 69.4%, with an overall test churn rate of 26.5%. What this means is that this model concentrated likely churners into a group that was about 2.6 times more likely to churn than the average test customer.

The final test AUC after running the final model for logistic regression was 0.847, and compared to the slightly higher 0.850 from boosted trees. Logistic regression produced the same 69.4% contact-list churn rate, which means boosted trees performed very slightly better overall. This is why I recommend treating boosted trees as a pilot candidate, while keeping logistic regression as a good backup and alternative reference.

One main limitation is that predicting churn does not necessarily prove that a retention offer will cause customers to stay. At 10% save rate, the boosted trees campaign would lose about \$1,620 per 1,000 contacts. At 15% save rate, it would earn an estimated \$670. This means that the break-even save rate would be around 13.5%. So the recommendation could reverse if the actual offer performs below this 13.5% level.

Summit Telecom should test the offer with a randomized pilot. I think that customers in the boosted trees model's highest-risk group should be randomly divided into an offer group as well as a control group. Both of these groups should be followed for the same time period, their churn rates, contact costs, and retained-customer value should be compared. Again, I would also reconsider my recommendation if the offer failed to exceed the 13.5% break-even save rate or if a larger test showed that potentially the logistic regression could produce better contact-list results or higher profitability.